Day 3 Solutions:

1. Convert the Temperature

This problem asks us to convert a temperature from Celsius to both Kelvin and Fahrenheit scales. We need to apply the standard temperature conversion formulas and return both results in an array. The conversions are straightforward mathematical transformations using well-established constants.

Kelvin = Celsius + 273.15

Fahrenheit = Celsius * 1.80 + 32.00

Code:

```
class Solution(object):
    def convertTemperature(self, celsius):
        Kelvin = celsius + 273.15
        Fahrenheit = celsius * 1.80 + 32.00
        return [Kelvin, Fahrenheit]
```

Complexity:

- Time complexity: O(1)
  We perform only constant-time arithmetic operations regardless of input value.
- Space complexity: O(1)
  We use only constant space for the calculations and the result array.

2. Smallest Even Multiple

This problem asks us to find the smallest positive integer that is both even and a multiple of n. If n is already even, then n itself is the answer since it's both even and a multiple of itself. If n is odd, we need to find the smallest even multiple, which is 2n (since multiplying an odd number by 2 gives the first even multiple).

1. Determine if n is even or odd using modulo operation
2. Even case: If n is even, return n directly (it's already even and a multiple of itself)
3. Odd case: If n is odd, return 2n (the first even multiple of an odd number)
4. Mathematical insight: For odd numbers, 2n is always the smallest even multiple
5. Efficient calculation: No iteration needed, just conditional multiplication

Code:

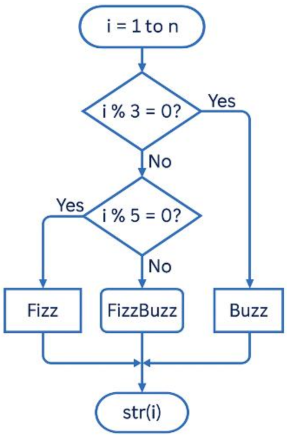
```
class Solution(object):
    def smallestEvenMultiple(self, n):
        if n % 2 == 0:
            return n
        else:
            return n*2
```

Complexity:

- Time complexity: O(1)
  We perform only constant-time operations: modulo check and potential multiplication.
- Space complexity: O(1)
  We use only constant extra space for the calculation.

3. Fizz Buzz

The classic FizzBuzz problem checks if numbers are divisible by 3, 5, or both. It’s often used to test clarity and simplicity in conditional logic.



FizzBuzz

Code:

```
class Solution:
    def fizzBuzz(self, n: int) -> List[str]:
        answer=[]
        for i in range(1,n+1):
            if i%3==0 and i%5==0:
                answer.append("FizzBuzz")
            elif i%3==0:
                answer.append("Fizz")
            elif i%5==0:
                answer.append("Buzz")
            else:
                answer.append(str(i))
        return answer
```

Complexity:

- Time Complexity: (O(n)) — loop runs n times.
- Space Complexity: (O(n)) — stores n results.